

Supplementary Material 2: Independent ANTs/MNI152 Registration-Pipeline Analysis in Symmetric MNI152 Space

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1 Purpose and scope

This analysis tests whether the study’s geometric and behavioral conclusions depend on the atlas-registration pipeline. It repeats deformation extraction, stationary-log/Helmholtz–Hodge analysis, descriptive and adjusted associations, and leakage-resistant nested cross-validation after independently registering the same lesion-filled anatomical images with ANTs. The input anatomy, lesion masks, inpainting targets, clinical table, outcomes, feature definitions, and statistical plan were held constant; registration software, reference atlas, and working resolution changed together. This is therefore a whole-pipeline sensitivity analysis, not an isolated ANTs-versus-SVReg software experiment. Agreement does not establish physical validity; disagreement identifies pipeline dependence.

2 ANTs registration to and from MNI152

The fixed reference was the 1-mm symmetric ICBM 152 nonlinear 2009a template (MNI152NLin2009aSym) and brain mask retrieved programmatically from TemplateFlow (Fonov et al., 2009; Ciric et al., 2022). Code verified canonical RAS orientation, orthogonal axes, an odd first dimension, a central voxel at world $x = 0$, and exact reflection symmetry of the brain mask. These checks are necessary because an array reversal is not a valid contralateral reflection in every nominal MNI geometry.

Each lesion-filled, brain-extracted T1-weighted image was resampled to 2-mm isotropic resolution and registered to the fixed reference with ANTsPy 0.6.1 using rigid, affine, and symmetric normalization (SyN) stages (Avants et al., 2008, 2011). Fixed and moving brain masks constrained all stages. The reproducible quick preset `antsRegistrationSyNQuickRepro[s]` used cross-correlation for the SyN stage, random seed 2026, single-precision arithmetic, and one ITK thread per case. Process-level parallelism did not change within-case threading or seed. The single SyN fit supplied forward and inverse mappings. Subject-to-MNI image resampling and MNI-to-subject image resampling were both saved outside the repository, together with the affine, forward warp, and inverse warp.

The quick reproducible schedule was selected after an engineering benchmark on one case. Relative to the full reproducible schedule, elapsed time decreased from 348.3 to 87.8 seconds, brain-mask Dice changed from 0.9012 to 0.9008, and intensity correlation changed from 0.5643 to 0.5493. This computational benchmark motivated a feasible complete-cohort rerun and is not cohort-level biological evidence.

ANTs applies transformations to images by pull resampling, whereas its point interface uses the forward list to map fixed points to moving points. We therefore transformed every fixed-brain voxel center through the forward list to construct the fixed-to-moving inverse coordinate map required by the main analysis. A deterministic sample of approximately 2,000 points was then mapped back with the inverse list. Registration QC required fixed-warped-moving brain Dice ≥ 0.70 , round-trip RMSE ≤ 0.50 mm, and a nonpositive-Jacobian fraction ≤ 0.001 in the raw SyN warp.

3 Deformation and log-domain analysis

The fixed-to-moving coordinate map was analyzed with the same contralesional-affine removal, mirrored homolog subtraction, exclusion masks, 2-mm physical smoothing, 3–20-mm shell, and laterality criterion as the main analysis. The 1,000-voxel support criterion of the 1-mm analysis was expressed as its resolution-invariant physical equivalent of 1,000 mm³, or 125 voxels on the 2-mm grid. Affine-fit residuals were converted from 2-mm source voxels to millimetres before reporting.

Stationary-log and Helmholtz–Hodge settings preserved the physical construction of the main analysis: a 2-voxel stride produced a 4-mm analysis grid; taper, smoothing, padding, exponentiation, and reconstruction criteria were unchanged. The representation remains an arbitrary-unit stationary registration parameter, not measured tissue velocity. No pressure, force, or tissue-mechanics parameter was estimated.

The same clinical and conventional lesion predictors, uncertainty features, ridge-penalty grid, 20 repeated nested five-fold splits, bootstrap sample count, association families, multiplicity corrections, and left-lesion-only analysis were regenerated. Because the retained participants and deterministic ordering matched the SVReg analysis, the common predictive models used identical outer splits. ANTs–SVReg descriptor agreement used Spearman correlation with participant-bootstrap 95% intervals. Direct algorithmic prediction comparisons used participant mean absolute error averaged over repeats; positive ANTs-minus-SVReg values favor SVReg.

ANTs registration did not include the direct non-inpainted SVReg rerun. The two descriptive correlations involving that SVReg-only sensitivity quantity were therefore omitted rather than represented by artificial zeros; Holm correction covered the ten measured ANTs descriptive associations. The six adjusted associations were unchanged in definition.

4 Registration and field quality control

The ANTs manifest contained 214 cases. 214 passed registration QC and 213 passed the combined registration/deformation criteria. Median brain-mask Dice was 0.971. Median round-trip RMSE was 0.075 mm (maximum 0.094 mm), and the maximum raw-warp nonpositive-Jacobian fraction was 0.000%. All 214 extracted stationary-log embeddings passed log-domain QC. Complete distributions are in Table A1 and Figure A1.

Table A1: Complete ANTs registration and deformation quality-control summaries.

Metric	Min	Q1	Median	Q3	Max
Brain-mask Dice	0.8668	0.9631	0.9715	0.9764	0.9830
Cycle RMSE (mm)	0.0502	0.0701	0.0749	0.0795	0.0939
Cycle maximum (mm)	0.2925	0.4147	0.4831	0.5538	0.8659
Intensity Pearson r	0.4720	0.6547	0.6995	0.7236	0.7791
Raw minimum Jacobian	0.1158	0.2719	0.3111	0.3467	0.4450
Raw folding fraction	0.0000	0.0000	0.0000	0.0000	0.0000
Normalized-field folding fraction	0.0017	0.0062	0.0077	0.0096	0.0218
Affine-fit RMSE (mm)	2.2053	2.8571	3.0364	3.2995	4.5001

5 Cross-algorithm descriptor agreement

Spearman rank agreement between ANTs and SVReg was $\rho = 0.178$ for median near-lesion displacement magnitude, $\rho = 0.130$ for stationary-log RMS, $\rho = 0.257$ for curl-free energy fraction, and $\rho = 0.259$

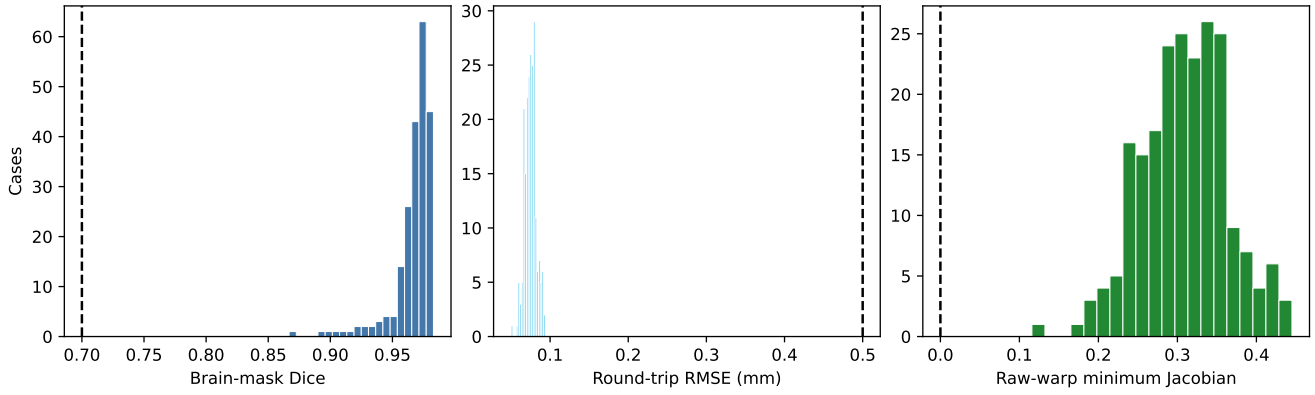


Figure A1: **ANTs registration quality control.** Dashed lines show the predefined eligibility boundaries. The Jacobian boundary is zero.

for divergence-free energy fraction. Table A2 reports all displacement and Hodge descriptors, including bootstrap intervals. Rank agreement quantifies pipeline sensitivity; it is not an equivalence test.

Table A2: ANTs-SVReg descriptor rank agreement on common QC-passing cases.

Family	Descriptor	n	ρ	CI low	CI high
Displacement	Median magnitude	213	0.178	0.040	0.308
Displacement	95th-percentile magnitude	213	0.358	0.228	0.479
Displacement	Median signed radial	213	0.664	0.571	0.744
Displacement	Mean absolute radial	213	0.287	0.155	0.413
Displacement	Outward radial integral	213	0.769	0.699	0.825
Displacement	Inward radial integral	213	0.677	0.590	0.748
Displacement	Log-Jacobian expansion integral	213	0.689	0.598	0.763
Displacement	Log-Jacobian compression integral	213	0.842	0.786	0.884
Hodge	Log-velocity RMS	214	0.130	-0.008	0.256
Hodge	Curl-free energy fraction	214	0.257	0.124	0.376
Hodge	Divergence-free energy fraction	214	0.259	0.129	0.388

6 Complete ANTs associations, including nonsignificant results

Table A3 reports the complete prespecified unadjusted association family. Table A4 reports all adjusted associations after rank residualization on the conventional clinical and lesion variables. The adjusted magnitude association was partial $\rho = -0.145$ (Holm $p = 0.182$), the curl-free association was $\rho = -0.131$ (Holm $p = 0.182$), and the divergence-free association was $\rho = 0.141$ (Holm $p = 0.182$).

7 Complete predictive results and direct algorithm comparison

The ANTs analysis cohort contained 210 participants. Adding ANTs deformation features to the conventional lesion model produced a mean MAE advantage of 0.81 points (95% interval 0.004 to 1.608; Holm $p = 0.038$). Adding ANTs Hodge features produced an advantage of -0.12 points (-0.53 to 0.30; Holm $p = 0.507$). Tables A5 and A6 retain every evaluated model and comparison, including nonsignificant findings.

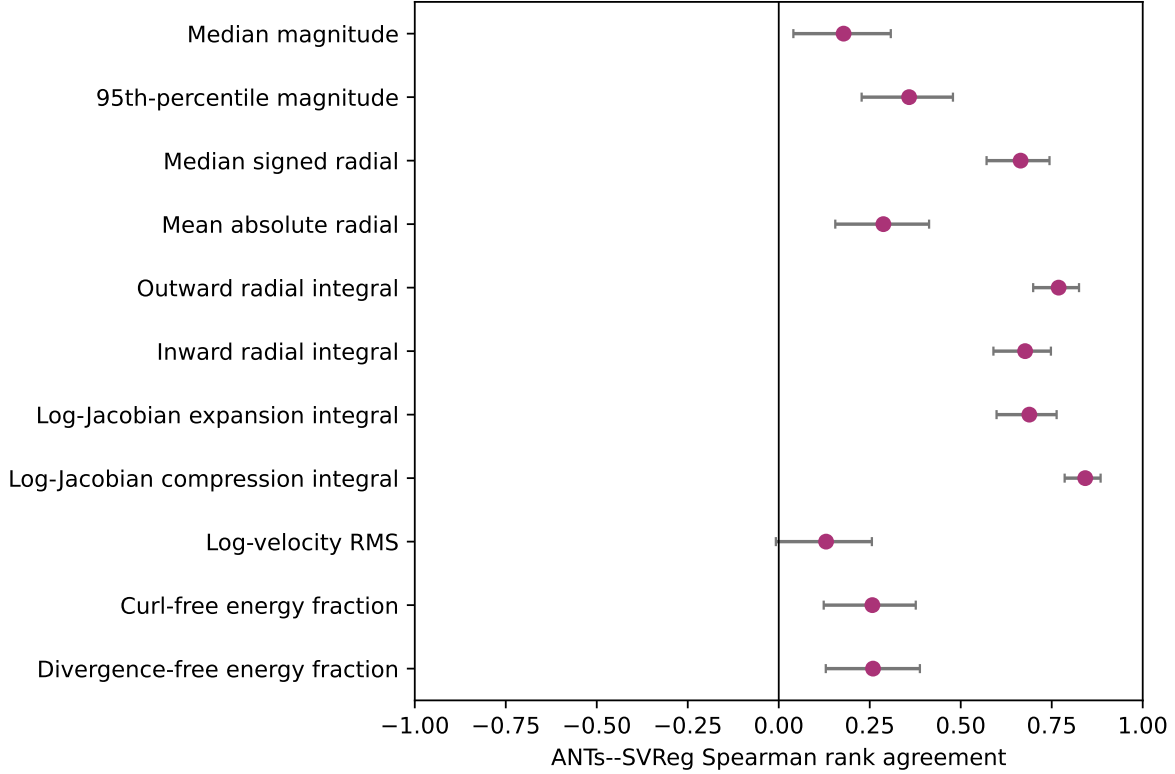


Figure A2: **ANTs–SVReg rank agreement.** Points are Spearman correlations; bars are participant-bootstrap 95% intervals.

On identical participants and splits, ANTs-minus-SVReg mean MAE was -0.44 points for the deformation-augmented model (95% interval -1.20 to 0.27), 0.06 for the Hodge-augmented model (-0.41 to 0.55), and -0.34 for the combined model (-1.02 to 0.32). Table A7 reports all common models; the shared conventional model is included as a zero-difference implementation check.

8 Interpretation and limitations

This sensitivity analysis changes the nonlinear registration implementation, reference atlas, and working resolution, but not lesion delineation, inpainting, behavioral measurements, or conventional lesion predictors. It therefore addresses whole-pipeline dependence, not an isolated software effect or all sources of measurement uncertainty. Agreement may remain inflated by shared input preprocessing, while disagreement can arise from reference anatomy, optimizer, similarity metric, regularization, or resolution. The 2-mm registration grid and reproducible quick schedule prioritize a feasible, fully regenerated cohort analysis; they do not reproduce the precise scale of SVReg optimization. Finally, neither a diffeomorphic ANTs warp nor a stable Hodge component supplies the constitutive parameters and boundary conditions required for pressure inference.

The within-ANTs deformation comparison met its prespecified criterion, but its lower bootstrap bound was close to zero, the corresponding SVReg comparison was inconclusive, and the direct same-split ANTs–SVReg MAE interval included zero. The appropriate interpretation is pipeline-sensitive evidence rather than a replicated deformation biomarker or a demonstrated advantage of one registration package.

Table A3: Complete ANTs descriptive Spearman association family.

Association	n	ρ	CI low	CI high	Holm p
Lesion volume vs AQ	210	-0.698	-0.769	-0.612	<0.001
Magnitude vs lesion volume	210	0.139	-0.004	0.274	0.045
Absolute radial vs lesion volume	210	0.291	0.157	0.416	<0.001
Magnitude vs AQ	210	-0.184	-0.316	-0.048	0.022
Absolute radial vs AQ	210	-0.270	-0.394	-0.134	<0.001
Signed radial vs AQ	210	-0.277	-0.395	-0.149	<0.001
Log-velocity RMS vs lesion volume	210	-0.462	-0.567	-0.345	<0.001
Log-velocity RMS vs AQ	210	0.377	0.255	0.483	<0.001
Curl-free fraction vs AQ	210	-0.212	-0.340	-0.078	0.008
Divergence-free fraction vs AQ	210	0.168	0.033	0.294	0.030

Table A4: Complete ANTs conventional-feature-adjusted association family.

Association	n	Partial ρ	CI low	CI high	Holm p
Magnitude vs AQ	210	-0.145	-0.271	-0.012	0.182
Absolute radial vs AQ	210	-0.118	-0.248	0.020	0.182
Signed radial vs AQ	210	-0.152	-0.282	-0.016	0.149
Log-velocity RMS vs AQ	210	0.069	-0.064	0.207	0.315
Curl-free fraction vs AQ	210	-0.131	-0.272	0.023	0.182
Divergence-free fraction vs AQ	210	0.141	-0.010	0.280	0.182

References

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Table A5: Complete repeated nested-CV results using ANTs-derived features.

Model	n	MAE	RMSE	R^2	r
Intercept only	210	24.14	27.68	-0.009	-0.119
Clinical	210	24.23	27.83	-0.020	-0.132
Conventional lesion	210	16.42	20.36	0.454	0.675
Lesion + deformation	210	15.61	19.55	0.497	0.705
Lesion + uncertainty	210	15.71	20.11	0.468	0.688
Uncertainty + deformation	210	15.29	19.28	0.510	0.715
Lesion + log-velocity Hodge	210	16.54	20.45	0.449	0.672
Deformation + log-velocity Hodge	210	15.78	19.65	0.492	0.702

Table A6: Complete paired MAE comparisons using ANTs-derived features.

Reference	Comparison	Advantage	CI low	CI high	Holm p
Conventional lesion	Clinical	-7.81	-9.88	-5.71	<0.001
Conventional lesion	Lesion + deformation	0.81	0.004	1.61	0.038
Conventional lesion	Lesion + uncertainty	0.71	-0.09	1.50	0.008
Conventional lesion	Uncertainty + deformation	1.13	0.35	1.91	0.005
Conventional lesion	Intercept only	-7.72	-9.73	-5.64	<0.001
Conventional lesion	Lesion + log-velocity Hodge	-0.12	-0.53	0.30	0.507
Conventional lesion	Deformation + log-velocity Hodge	0.64	-0.18	1.47	0.114
Lesion + uncertainty	Uncertainty + deformation	0.42	-0.24	1.10	0.162
Lesion + deformation	Deformation + log-velocity Hodge	-0.17	-0.47	0.11	0.292

Table A7: Direct ANTs-SVReg prediction comparison on shared outer-CV splits. Positive Δ MAE indicates larger error with ANTs.

Model	SVReg MAE	ANTs MAE	Δ MAE	CI low	CI high	p
Clinical	24.23	24.23	0.00	0.00	0.00	1.000
Intercept only	24.14	24.14	0.00	0.00	0.00	1.000
Lesion + log-velocity Hodge	16.48	16.54	0.06	-0.41	0.55	0.590
Lesion + deformation	16.04	15.61	-0.44	-1.20	0.27	0.681
Deformation + log-velocity Hodge	16.12	15.78	-0.34	-1.02	0.32	0.628
Lesion + uncertainty	15.71	15.71	0.00	0.00	0.00	1.000
Conventional lesion	16.42	16.42	0.00	0.00	0.00	1.000
Uncertainty + deformation	15.77	15.29	-0.48	-1.07	0.05	0.228